

HOMework 3

NUMERICAL ANALYSIS

(MATH 461) FALL, 2023

DUE: 09/25/2023

Problem 1. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a twice continuously-differentiable function with $f''(x) > 0$ for all x , and let x^* be a point such that $f(x^*) = 0$ and $f'(x^*) < 0$. Suppose the secant method is applied with starting values x_0 and x_1 to generate the iterates x_k .

- (a) Show that if $x_0, x_1 < x^*$, then $x_2 < x^*$ as well.
- (b) Find analogous results for $x_0, x_1 > x^*$ and when x_0, x_1 are on opposite sides of x^* . You do not need to be rigorous if the claim is obvious from a sketch.

Problem 2. Suppose a method is used to estimate a solution $x^* = 0$ to $f(x) = 0$ by generating the sequence of approximations $x_k = 0.99^k$ (which, one can see, does indeed converge to x^*). How many iterations are required to reach an error less than 10^{-6} ? If the iteration is stopped when $|x_{n+1} - x_n| < 10^{-6}$, how many iterations fewer than necessary were computed?

Problem 3. Suppose $f \in C^{m+1}(\mathbb{R})$ for some $m > 1$, suppose $|f^{(m+1)}(x)| \leq C$ for some $C \in \mathbb{R}$ and all $x \in \mathbb{R}$, and suppose x^* is a root of f with multiplicity m (i.e. $f(x^*) = f'(x^*) = \dots = f^{(m-1)}(x^*) = 0$ and $f^{(m)}(x^*) \neq 0$). Consider the iteration

$$x_{k+1} = x_k - m \frac{f(x_k)}{f'(x_k)}$$

and assume $x_k \rightarrow x^*$ as $k \rightarrow \infty$. Show that the convergence of x_k to x^* is superlinear.

Hint: Consider the Taylor expansions

$$f(x_k) = f(x^*) + f'(x^*)(x_k - x^*) + \dots + \frac{1}{m!} f^{(m)}(x^*)(x_k - x^*)^m + \frac{1}{(m+1)!} f^{(m+1)}(\zeta_k)(x_k - x^*)^{m+1}$$

$$f'(x_k) = f'(x^*) + f^{(2)}(x^*)(x_k - x^*) + \dots + \frac{1}{(m-1)!} f^{(m)}(x^*)(x_k - x^*)^{m-1} + \frac{1}{m!} f^{(m+1)}(\xi_k)(x_k - x^*)^m$$

where ζ_k, ξ_k are points between x^* and x_k .

Problem 4. Let A and B be lower triangular matrices. Prove that $C = AB$ is also lower triangular.

Problem 5. Let A be an $n \times n$ matrix, and let the linear transformation $T : \mathbb{R}^n \rightarrow \mathbb{R}^n$ be defined by $T(x) = Ax$. Suppose $T(u) = T(v)$ for some $u, v \in \mathbb{R}^n$ with $u \neq v$.

- (a) Does the equation $Ax = 0$ have a nontrivial solution? Justify your answer.
- (b) Does the equation $Ax = z$ have a unique solution for any $z \in \mathbb{R}^n$? Justify your answer.
- (c) Briefly and informally explain why the answer to part (b) of this problem may be relevant when conducting numerical computations involving floating point numbers.